

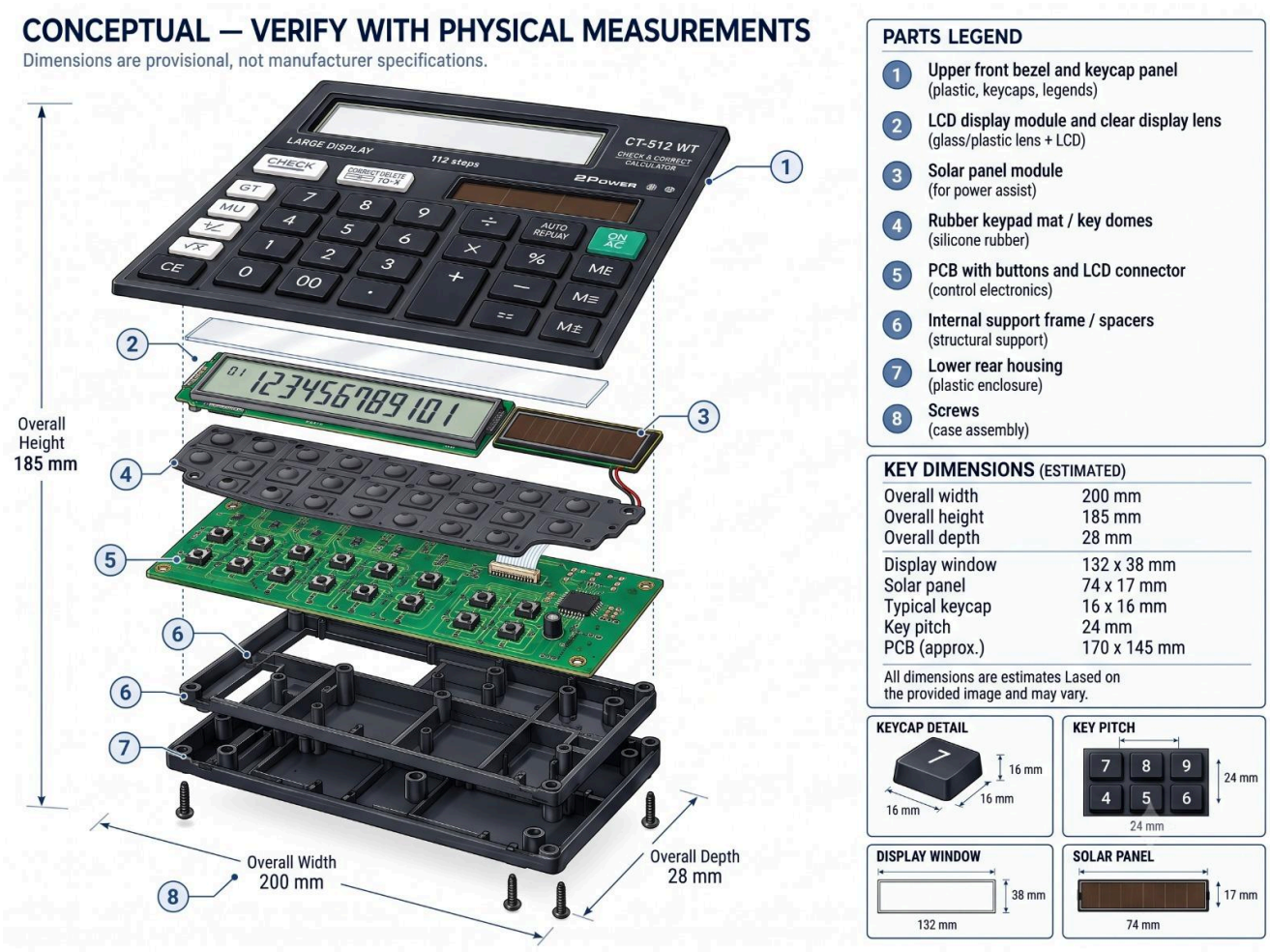

# REVERSE ENGINEERING

## DIGITAL CALCULATOR (MODEL: CT-512 WT)

### 01. EXTERNAL FEATURES

he CT-512 WT is a compact electronic calculator designed to perform basic mathematical operations quickly and accurately. It accepts numerical inputs through a keypad, processes them using an electronic computing circuit, and displays the results on an LCD.

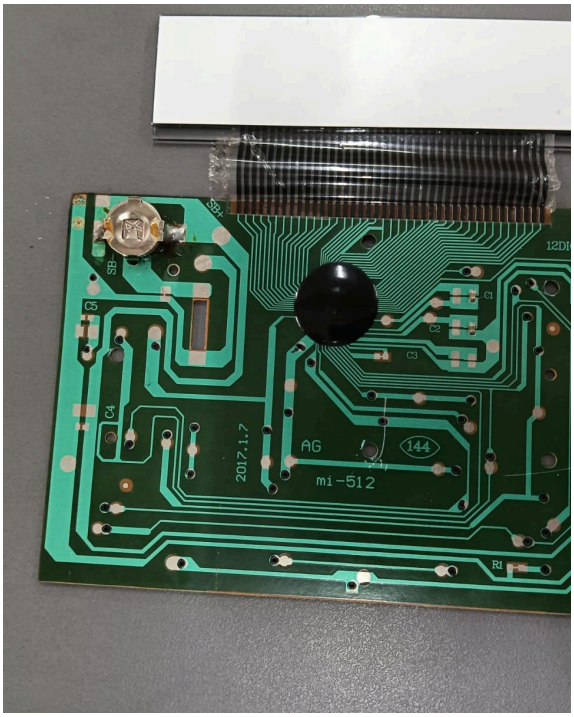
It is commonly used for everyday calculations in educational, domestic, and office environments.




# 02. PHYSICAL CONSTRUCTS

a. Compact plastic body
b. Large LCD display
c. Numeric keypad and function keys
d. Lightweight, portable design
e. Printed circuit board (PCB) inside






03.

# EXTERNAL SUBSYSTEMS

a. Input subsystem: keypad
b. Processing subsystem: calculator IC
c. Output subsystem: LCD
d. Power subsystem: battery and, if confirmed on your model, a solar cell



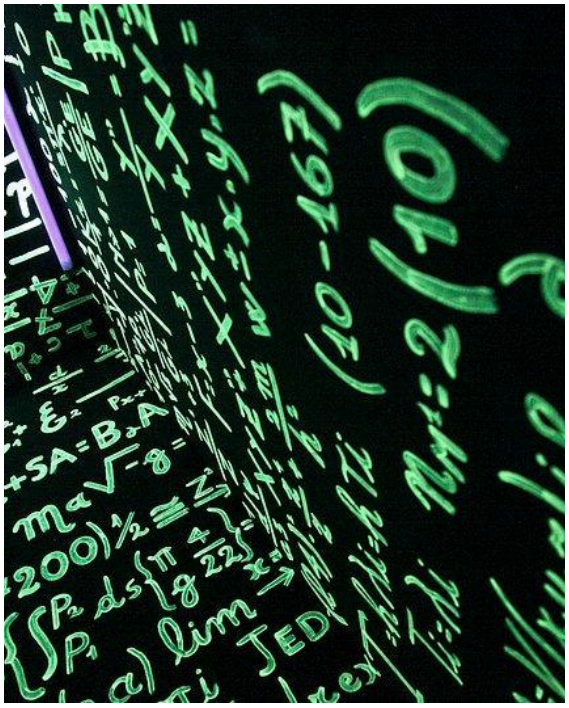

# REVERSE ENGINEERING

## DIGITAL CALCULATOR (MODEL: CT-512 WT)

## 04. DESCRIPTION OF THE SUB SYSTEMS

| SUBSYSTEMS | DESCRIPTION |
|---|---|
| Keypad | Accepts numbers and mathematical commands |
| LCD | Displays numbers, operators and results |
| Power Supply | Provides the electrical energy required |
| PCB | Connects and supports the electronic components |
| | |

# 05. MODES OF OPERATION



f. Basic arithmetic: Addition, subtraction, multiplication and division.
g. Percentage: Calculates percentages.
h. Memory: Stores and recalls numerical values.
i. Clear functions: Clears the current entry or resets the calculation.

j. Decimal calculations: Performs calculations involving decimal numbers.




## 06. PURPOSE OF THE PRODUCT

e. Reduce the time required for manual calculations.
f. Minimize arithmetic errors.
g. Provide an easy-to-use interface.
h. Offer a portable and low-power computing solution.

## 07. USER NEEDS ADDRESSED

a. Quick calculation of mathematical expressions.
b. Simple operation through dedicated keys.
c. Clear and readable numerical output.
d. Portability for use in classrooms, homes, and offices.
e. Low power consumption and reliable operation.

# PRODUCT ORIGIN TO LATEST INVENTIONS

## Evolution of Calculators

***Use the following stages to explain how calculators have developed over time.***

1. Early stage — Mechanical calculators

Calculations were performed using gears, wheels, and mechanical parts.

Limitation: Bulky construction and limited functionality.

2. First generation — Early electronic calculators

Electronic circuits replaced mechanical calculation mechanisms. Early models commonly used LED displays.

Improvement: Faster calculations and electronic operation.

3. Second generation — Compact LCD calculators

Integrated circuits and LCD technology made calculators smaller and more energy-efficient.

Improvement: Lower power consumption and longer battery life.

4. Latest generation — Advanced calculators

Scientific, programmable, and graphing calculators support advanced mathematical operations.

Improvement: More functions, greater computing capability, and advanced displays.

# Current Status — Core Smart Technologies Used in the Product

*The CT-512 WT uses established electronic technologies to provide a simple, reliable calculation system.*

- *Integrated circuit (IC): Performs the calculation and control operations.*
- *LCD technology: Displays the calculated values using low power.*
- *Key scanning: Detects the key pressed by the user.*
- *CMOS technology: Commonly used in low-power calculator ICs.*
- *PCB technology: Provides electrical connections between components.*
- *Battery-powered operation: Enables portable use.*

*Note: The exact IC architecture and semiconductor process used in your particular calculator cannot be confirmed from the external images alone.*

# Engineering Importance of This Evolution

The evolution of calculators demonstrates how electronic engineering has improved computing devices.

1. Miniaturization has reduced the size of electronic circuits.
2. Integrated circuits have reduced component count and complexity.
3. LCDs have significantly reduced display power consumption.
4. Improved semiconductor technology has increased reliability.
5. Low-power designs have enabled compact, portable devices.

# INTERNAL COMPONENTS & WORKING FLOW

### *Introduction to the Internal Design*

The calculator consists of a keypad, a calculator IC, an LCD, a power supply, and a printed circuit board.

When a key is pressed, the input is detected by the calculator IC. The IC processes the input, performs the required mathematical operation, and sends the appropriate signals to the LCD.

The PCB provides the electrical connections between these subsystems.

## Photograph of Internal Components

## Components Present Inside Calculator

| COMPONENT | FUNCTION |
|---|---|
| Black epoxy-covered IC | Main calculator processing circuit |
| PCB tracks | Electrical connections between components |
| LCD connector | Transfers display signals |
| Resistors | Provide resistance and support circuit functions |
| Capacitors | Provide filtering or other supporting functions |
| Switch contacts | Detect keypad inputs |
| Battery contact/holder | Supplies electrical power |

The black epoxy-covered IC is commonly called a glob-top package. The silicon die is concealed beneath the epoxy, so its exact part number cannot be read directly from the photograph.

## Internal Block Diagram (Functional View)

## Internal Subsystems with Detailed Explanation

1. Input subsystem — Keypad

- Function: Accepts numerical and mathematical inputs.
- Working principle: Pressing a key changes the electrical connection between keypad contacts.

- Characteristics: Multiple keys for digits, arithmetic operations, and control functions.
- Role: Provides the user interface.

**2. Processing subsystem — Calculator IC**

- Function: Processes inputs and performs calculations.
- Working principle: Detects key inputs, stores relevant values, executes arithmetic operations, and generates display-control signals.
- Characteristics: Compact, low-power integrated electronic circuit.
- Role: Acts as the brain of the calculator.

**3. Output subsystem — LCD**

- Function: Displays numbers and calculated results.
- Working principle: Electrically controlled liquid-crystal segments form the required digits.
- Characteristics: Low power consumption and good readability.
- Role: Communicates the result to the user.

**4. Power subsystem**

- Function: Supplies the required operating voltage.
- Working principle: A battery, and a solar cell if present, provides energy to the circuit.
- Characteristics: Designed for low-power operation.
- Role: Enables portable use.

**5. PCB subsystem**

- Function: Connects the electronic components.
- Working principle: Copper tracks carry electrical signals and power between circuit elements.
- Characteristics: Compact layout with interconnected conductive paths.
- Role: Provides the physical and electrical foundation of the device.

## Complete Working Flow (Step-by-Step)

1. Power ON

The power supply energizes the calculator circuit.

2. Key Press

The user enters a number or mathematical operation.

3. Input Detection

The calculator IC identifies the pressed key.

4. Data Processing

The IC stores the input and performs the required calculation.

5. Display Conversion

The IC generates the appropriate LCD control signals.

6. Result Display

The LCD displays the calculated value.

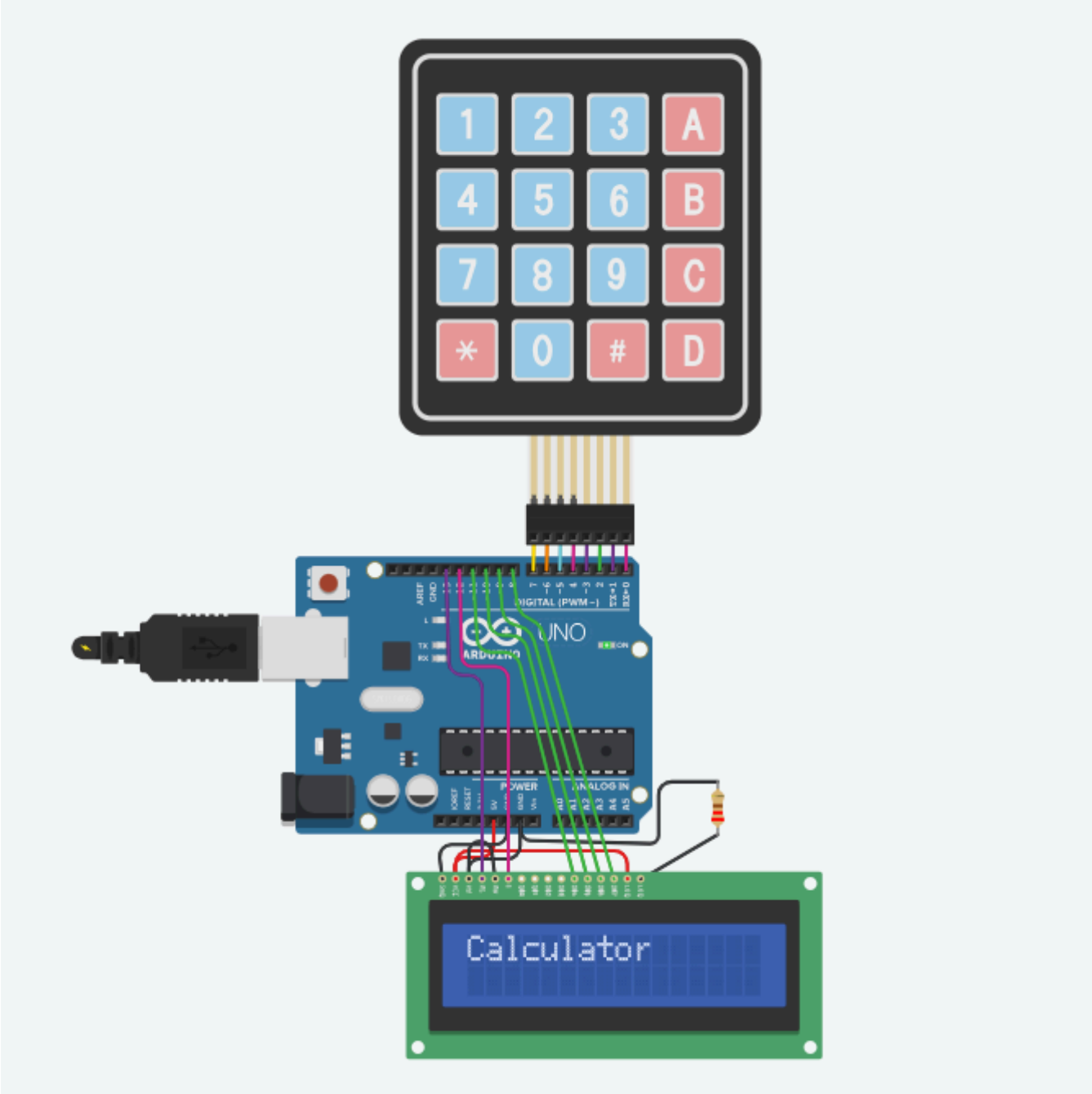
Component List — Bill of Materials (BOM)

| S. No. | Component | Quantity |
|---|---|---|
| 1 | Calculator IC | 1 |
| 2 | LCD display | 1 |
| 3 | Printed circuit board | 1 |
| 4 | Keypad membrane/contact assembly | 1 set |

| | | |
|---|---|---|
| 5 | Resistors | As fitted |
| 6 | Capacitors | As fitted |
| 7 | Battery and contacts | As fitted |
| 8 | Plastic casing | 1 set |
| 9 | Solar cell | If present |
| 10 | LCD connector/contact strip | 1 set |

# SIMULATION & RESULTS

## Circuit Diagram of the Product




# Simulation Circuit

https://www.tinkercad.com/things/cBiFAMtwcdc-digital-calculator/editel?returnTo=https%3A%2F%2Fwww.tinkercad.com%2Fdashboard%2Fd

esigns%2Fall&sharecode=JcKZUVgbHomzbAJoNf9dPlzYlEJxCbbHOOUjfi58Ch4

## Simulation Results

| INPUT | OPERATION | EXPECTED OUTPUT |
|---|---|---|
| 25,15 | Addition | 40 |
| 25,15 | Subtraction | 10 |
| 5,6 | Multiplication | 30 |
| 20,4 | Division | 5 |
| 200,10 | Percentage | 20 |

*Note the methods used to gather data for the case study*

Many case studies use a mixed-methods approach that combines qualitative and quantitative data collection techniques. The level of detail in this section depends on the audience for the case study.

# 01. Approaches used

*Outline the specific approaches or frameworks used*

Framework could include theoretical models, industry standards, or best practices. Theoretical models help people understand the situation. Industry standards and best practices can help evaluate performance, gain insights, and identify areas to improve.

# 02. Relevant facts and information

*Present key facts*

a. In a few bullet points, present key facts, figures, and information that surfaced during analysis, such as: project specifications, timelines, budgets, or other quantitative data
b. Describe how those facts relate to the problem statement
c. Include specific sources and data points where possible

# Conclusion

The reverse engineering of the CT-512 WT calculator helped us understand the integration of input, processing, output, and power subsystems in a compact electronic device.

By studying its external features, internal PCB, components, and working principles, we explored the practical application of electronic circuits and integrated circuit technology.

This project demonstrates how a simple consumer electronic product combines hardware design, semiconductor technology, and digital computation to perform useful real-world tasks.